

Distributed CDN System – Demonstration Guide (Final)

1. Start Backend Services

```
cd distributed-cdn-system
docker-compose down
docker-compose up --build -d
docker ps
```

2. Service Verification

```
curl http://localhost:8000/health
curl http://localhost:8005/nodes
curl http://localhost:8001/health
curl http://localhost:8002/health
curl http://localhost:8003/health
```

3. Basic Routing (with location header)

```
curl -H "X-Client-Location: Asia" "http://localhost:8000/route?file=test.txt"
curl -H "X-Client-Location: Asia" "http://localhost:8000/route?file=test.txt"
```

4. Multi-Region Routing

```
curl -H "X-Client-Location: America" "http://localhost:8000/route?file=test.txt"
curl -H "X-Client-Location: Europe" "http://localhost:8000/route?file=test.txt"
```

5. Upload File to Origin

```
curl -X POST http://localhost:8004/upload \
-F "file=@new_test.txt"
```

6. Edge Fetch (MISS then HIT)

```
curl "http://localhost:8002/fetch?file=new_test.txt"
curl "http://localhost:8002/fetch?file=new_test.txt"
```

7. Cache Invalidation (Purge)

```
curl -X POST http://localhost:8006/purge \
-H "Content-Type: application/json" \
-d '{"file": "new_test.txt"}'
```

8. Load Test (Concurrent Requests)

```
for i in {1..10}; do curl -H "X-Client-Location: Asia" "http://localhost:8000/route?file=test.txt" & done
```

9. Failover Test

```
docker stop distributed-cdn-system-edge-a-1
curl -H "X-Client-Location: Asia" "http://localhost:8000/route?file=test.txt"
docker start distributed-cdn-system-edge-a-1
```

10. Optional Edge Failure APIs (if implemented)

```
curl -X POST http://localhost:8003/fail
curl -X POST http://localhost:8003/recover
```

11. Logs (Optional)

```
docker-compose logs -f
```

12. Frontend

```
cd cdnFrontend
npm install
npm run dev
Open: http://localhost:3000
```

13. Demonstration Flow

1. Upload file to origin 2. Route request → MISS 3. Repeat request → HIT 4. Purge cache → MISS again 5. Demonstrate failover 6. Show frontend interaction